

INTERNATIONAL FALLS, MN, USA

WMO: 727470

Lat: 48.561N

Lon: 93.398W

Elev: 1183

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-20.8	-30.4	1.0	-24.5	-25.9	1.4	-19.4	20.8	12.4	19.1	10.2	3.6	290	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	85.1	69.8	82.2	67.7	79.5	66.0	72.8	81.4	70.6	79.0	68.5	76.1	9.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	114.4	78.1	67.6	106.0	75.2	65.6	98.6	73.1	37.2	80.9	35.2	79.2	33.5	76.0	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	18.0	16.3	-34.8	90.0	7.0	3.0	-39.9	92.2	-44.0	94.0	-47.9	95.7	-53.0	97.9		
			-31.8	76.2	4.8	2.1	-35.2	77.7	-38.0	78.9	-40.7	80.1	-44.2	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	5.5	10.1	24.1	38.6	51.3	61.3	65.6	63.2	55.2	42.2	27.0	12.3	
	DBStd	23.43	15.04	14.08	13.63	9.74	8.61	6.97	5.75	6.06	8.34	8.75	11.42	13.76	
	HDD50	5925	1380	1117	804	356	88	4	0	1	40	274	691	1170	
	HDD65	9984	1845	1537	1267	791	432	150	64	107	309	707	1140	1635	
	CDD50	1611	0	0	2	15	128	343	482	410	197	33	1	0	
	CDD65	195	0	0	0	0	7	38	82	51	16	1	0	0	
	CDH74	2333	0	0	1	8	152	481	907	596	178	10	0	0	
	CDH80	507	0	0	0	1	32	112	206	124	31	1	0	0	
Wind	WSAvg	7.4	7.4	7.7	8.0	8.3	8.2	7.1	6.4	6.3	6.9	7.8	7.8	7.2	
Precipitation	PrecAvg	24.7	0.8	0.6	1.0	1.6	2.8	3.9	3.7	2.9	3.0	2.0	1.3	0.9	
	PrecMax	32.8	3.7	1.6	3.7	3.6	6.7	10.2	9.5	6.7	7.4	4.8	3.5	2.2	
	PrecMin	16.9	0.0	0.0	0.1	0.1	0.2	0.7	1.1	0.6	0.3	0.2	0.0	0.2	
	PrecStd	3.7	0.7	0.4	0.7	0.9	1.5	1.9	2.0	1.3	1.7	1.2	0.8	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	45.3	63.9	73.2	84.4	89.0	88.4	87.1	83.9	74.9	59.9	41.8	
		MCWB	34.9	39.4	51.3	56.4	63.3	72.6	72.6	71.8	68.6	60.1	50.3	36.2	
	2%	DB	34.3	39.7	53.0	67.0	78.5	82.7	85.0	83.5	79.2	67.7	52.1	36.5	
		MCWB	32.1	35.4	44.8	50.5	60.4	67.2	70.4	69.4	65.9	56.4	46.0	33.6	
	5%	DB	30.8	35.2	47.0	61.6	73.1	79.3	82.1	80.4	74.8	62.2	46.7	33.6	
		MCWB	28.8	32.4	39.6	47.5	57.0	64.8	68.6	67.5	63.8	53.9	41.7	31.8	
	10%	DB	27.1	31.4	41.9	56.3	68.6	76.0	79.3	77.2	70.8	56.8	42.2	30.9	
		MCWB	25.5	28.8	36.3	44.9	55.2	63.3	66.9	65.6	62.0	49.8	38.4	29.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	40.2	54.1	57.7	67.4	74.3	75.6	74.7	71.6	62.6	52.3	37.0
			MCDB	38.7	43.7	58.5	68.2	77.4	84.9	85.2	83.6	80.7	69.2	57.5	39.9
2%		WB	32.6	35.4	45.4	52.1	63.5	70.5	73.1	71.9	68.4	58.4	46.3	34.1	
		MCDB	34.6	38.5	52.9	63.7	72.7	80.0	80.9	80.7	75.4	66.8	51.4	36.0	
5%		WB	28.9	32.6	40.4	48.7	59.9	67.9	70.9	69.4	65.7	54.3	42.3	32.0	
		MCDB	30.7	35.3	46.0	59.7	69.5	75.1	78.9	77.2	72.9	60.9	46.3	33.8	
10%		WB	25.6	28.8	36.7	45.6	57.0	65.3	68.8	67.4	62.8	50.6	38.4	29.1	
		MCDB	27.2	31.4	41.4	55.9	66.5	72.8	76.5	74.6	69.1	55.8	42.0	30.9	
Mean Daily Temperature Range	5% DB	MDBR	19.3	21.7	21.3	21.8	23.6	22.6	23.3	23.7	22.1	17.8	14.7	16.2	
		MCDBR	17.9	20.9	24.4	32.1	32.2	27.7	26.7	27.1	27.8	26.9	21.2	16.5	
		MCWBR	15.6	16.7	16.3	19.0	16.6	14.5	14.0	14.6	16.0	16.7	14.9	14.2	
	5% WB	MCDBR	16.7	18.6	22.4	28.5	26.0	22.7	22.9	23.4	22.6	23.9	19.4	15.7	
		MCWBR	14.8	15.0	15.8	17.7	14.8	13.2	13.6	13.9	14.2	15.7	14.4	13.8	
Clear-Sky Solar Irradiance	taub	0.250	0.268	0.308	0.339	0.355	0.373	0.384	0.372	0.354	0.305	0.284	0.254		
	taud	2.365	2.323	2.364	2.403	2.413	2.385	2.375	2.440	2.462	2.585	2.505	2.401		
	Ebn at Noon	266	285	288	289	287	282	276	275	270	271	252	248		
	Edh at Noon	21	28	32	34	35	37	36	33	29	22	19	18		
All-Sky Solar Radiation	RadAvg	411	792	1185	1516	1669	1766	1865	1579	1163	673	371	291		
	RadStd	54	82	109	135	110	140	107	109	97	92	47	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page